

INDUSTRIAL AUTOMATION IN THE NEW ECONOMY

The business landscape for manufacturers continues to shift significantly. But there is good news amid the volatility. Manufacturing convergence and the advanced technology that enables it offer the manufacturers an opportunity to make changes to bring their businesses in sync with the new market realities



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The global financial downturn has caused a dramatic decline in demand for consumer and capital goods. Manufacturers, in particular, have felt unprecedented distress. Yet there is good news: the innovative technology and sophisticated operational systems that manufacturers need to thrive in this new economy already exist.

Significant advances in automation technology, coupled with manufacturing convergence, can help manufactur-

ers optimise production and gain the efficiencies necessary for success in the new economy.

Industrial automation essentially involves leveraging the power of technology to reduce the dependency on human presence and decision making for any process. It leverages different electronic equipment (either standalone or interlinked with appropriate applications) to control different parameters of any industrial process.

Depending on the complexity of the process and type of equipment, current systems for process automation and control include distributed control sys-

tems, programmable logic controllers (PLCs), batch controllers, validation systems, ANSI/ISA S88.01, FieldBus, Foundation FieldBus, Profibus, AS-Interface, DeviceNet, PID controllers, human-machine interfaces (HMIs), OLE for process control and process instrumentation amongst others.

Advances in automation technology

There have been lots of advancements made in the areas of embedded computing, information technology and communication. Industrial automation companies have piggy-backed onto these advancements in the technology and come up with product lines that are capable of fast processing, compact in nature and open in architecture. The gap between the shop floor and the office floor is continuously being bridged through smart sensing, data acquisition and reporting.

Smart sensing devices with connectivity to smart inputs/outputs have given insights into valuable field and device information leading to improvement in productivity and asset maintenance.

PLCs and other controllers have become faster with high-speed processors. Standardised function blocks with faceplates for various applications have started making a lot of difference in engineering efforts.

On the technological front, the industry is gradually going towards nano-processing from microprocessing.